

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 21

Status: active BA5 execution plan after BA5-T2 canonical semantic registry coverage and governed-extension pressure testing.

Supersedes: Revision 20 only for forward execution state; R1-R20 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- BA2 relation/action vocabulary: CLOSED BY BA2-T4.
- BA3 provenance/derivation/identity/lifecycle/change-revalidation: CLOSED BY BA3-T4.
- BA4 projection boundary/traceability/coverage/interpretation: CLOSED BY BA4-T3.
- BA5-T1: COMPLETED / PROVISIONAL PASS WITH CANONICAL-REFERENT-NAMING LOWER-BOUND.
- BA5-T2: COMPLETED / PROVISIONAL PASS WITH DOMAIN-SCOPED REGISTRY AND GOVERNED-EXTENSION REFINEMENT.
- **BAReferent** and **BAProposition**: ACCEPTED first-class semantic identity families.
- ThreatForge remains a case study/tooling instrument, not DDTA semantic authority.

2. BA5 objective - canonical semantic registry and controlled authoring

BA5 uses a strict authoring hypothesis:

registered canonical token
-> normative semantic authoring

unregistered alias / synonym
-> not silently equivalent

The purpose is to make cross-document, Base Analysis and projection integration deterministic before considering lexical normalization as optional future assistance.

3. BA5-T1 result

T1 establishes one exact canonical referent name per governed baseline/naming scope, rejects aliases as second normative entity identifiers and preserves BA3 semantic identity across a governed rename.

`canonicalName != BAReferent identity`

Shared projections use the canonical referent name when presenting that shared entity; method-owned types/aggregations may retain local labels.

4. BA5-T2 result

T2 extends exact-token discipline across BA2 vocabulary and finds five refinements.

4.1 Domain-scoped registries

One flat global token namespace is rejected. Registry lookup preserves semantic domain and, for roles, operator context.

4.2 Exact BA2 keys

All thirteen operators remain exact canonical keys. Roles resolve as:

`(semanticOperatorKey, roleKey)`

4.3 Extensible domains

`semanticKind` and controlled reusable value domains may admit new method-neutral entries when evidence proves a genuine distinction. They do not require a universal pre-enumerated taxonomy.

Current T2 pressure explicitly covers:

```
store    contract
affirmative  negative
condition  temporalScope
ownership  management
```

`channel` remains DEFERRED / EVIDENCE-GATED rather than being promoted for convenience.

4.4 Immutable revision resolution

Mutable registry definitions are rejected. Extensible and methodology-contract registries resolve through immutable revisions; referent names remain anchored to governed documentation baselines/naming scopes.

4.5 Extension routing

```
alias -> reject/use existing key
method-specific -> keep downstream
needs independent identity -> BAReferent
genuine extensible-domain entry -> governed registry revision
genuine new operator/role contract -> smallest BA2 reopen
```

No normative synonym table is forced.

5. BA5-T3 - canonical registry, controlled-authoring and no-synonym closure review

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA5-T3 must perform one integrated adversarial closure review over T1 and T2.

It must pressure-test at least:

1. the complete registry-set boundary without collapsing all tokens into one namespace;
2. referent canonical-name consistency together with a governed cross-baseline rename;
3. immutable registry revision change while historical BA/projections remain reproducible;
4. exact operators and operator-scoped roles across facial and order/WMS examples;
5. extensible `semanticKind` / controlled-value admission versus closed BA2 contract changes;
6. a human projection plus two incompatible methodology projections without method taxonomy entering BA;
7. alias/synonym attempts at referent, operator, role, semantic-kind and controlled-value levels;
8. tool validation/completion/extension-request assistance without semantic-authority leakage;
9. whether any material counterexample forces synonym support; and
10. whether any BA5 responsibility forces the smallest BA1-BA4 reopen.

BA5-T3 guardrails

Do not:

- build a synonym ontology by default;
- perform arbitrary-narrative migration;
- authorize NLP/LLM output as semantic truth;
- define a universal semantic-kind taxonomy;
- promote method-specific terminology into Base Analysis;
- change closed BA2 operators/roles through a lexical registry extension;
- define ThreatForge classes/tables/UI;
- start BA6;
- define AnalysisRecord/Common Finding or formal STRIDE/STRIDE-AI schemas.

BA5-T3 exit condition

If no material counterexample survives, close BA5 for current thesis scope and freeze the smallest canonical-registry/controlled-authoring contract with explicit reopen criteria.

If a material counterexample survives, reopen only the smallest affected BA5 or earlier semantic responsibility.

6. BA6 - complete Base Analysis regression and closure

Only after BA5 closes, regress the complete BA0-BA5 design against the closed corpora and at least one structurally different holdout.

BA6 should emphasize the end-to-end path:

governed documentation

- > canonical registry-controlled authoring
- > accepted BA
- > provenance/change continuity
- > multiple projections
- > rebuild / cross-projection comparison

BA6 may reopen only the smallest earlier responsibility on a material counterexample.

Only BA6 may close the complete Base Analysis milestone for current thesis scope.

7. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

8. Next authorized microstep

Only after the BA5-T2 package is reviewed, committed, pushed and remotely verified, execute:

BA5-T3 - canonical registry, controlled-authoring and no-synonym closure review.

Do not start BA6 or downstream analysis schemas before BA5 is explicitly closed.